

R3: Longitudinal Beam Dynamics and Bunch Compression for the 0.5 ps and 2 ps Operating Modes

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1 Introduction

The UH MkV FEL is designed to operate at two bunch lengths: 2 ps (baseline) and 0.5 ps (short-pulse mode). A reviewer asks whether the transverse matching changes between modes and whether the transport line can compress a 2 ps beam to 0.5 ps.

This report consolidates the longitudinal studies (W9, W12) into a single document. We extract the full 6D transfer map from COSY INFINITY, propagate bunches at both operating modes, test $R_{56} = 0$ as an optimisation objective, evaluate in-line bunch compression through chirp, and validate against RF-Track particle tracking.

2 Transfer Map Analysis

2.1 COSY 6D Map Extraction

The 11-stage COSY FIT optimisation ($E = 40$ MeV, $\varepsilon_n = 8 \pi \cdot \text{mm} \cdot \text{mrad}$, FR0, order 3) produces $\text{MSE} = 2.27 \times 10^{-7}$. The full 6×6 linear transfer map $\mathcal{M}$ and the second-order coefficient $(l|\delta\delta) = \text{ME}(5, 66)$ are extracted via the COSY PM command.

COSY uses $\delta = \delta_K = \Delta K/K_0$ (kinetic energy deviation). The standard-optics conversion factor is $p_0 c/K_0 = 40.508/40 = 1.013$ for 40 MeV electrons.

Table 1 lists the longitudinal map elements at the undulator entrance.

The transfer map has the expected block structure:

- $(\delta|x_j) = 0$ for all transverse x_j : no transverse→energy coupling. The transverse and longitudinal subspaces are decoupled in the Twiss matching sense.
- $(l|x)$, $(l|a) \neq 0$: off-axis particles travel different path lengths through dipoles. This is a geometric effect that does not affect transverse matching.
- $(l|\delta) = 27.09$ mm: the chromatic path-length spread converts energy spread into bunch lengthening.
- $(l|\delta\delta) = 0$: no nonlinear compression or decompression effect.

3 6D Bunch Propagation

Gaussian bunches ($N = 10,000$, seed = 42) are propagated through the COSY map for four scenarios: 0.5 ps and 2 ps, each with $h = 0$ and $h = 5 \times 10^9 \text{ s}^{-1}$.

Table 1: Longitudinal elements of the COSY transfer map $\mathcal{M}$ at the undulator entrance. COSY notation $(y_i|y_j)$; transport-code equivalents (R_{ij}, T_{ijk}) for reference.

Map element	FOX	Transport	Value	Interpretation
$(l x)$	ME(5,1)	R_{51}	9.874×10^{-3}	Path length from x (dipoles)
$(l a)$	ME(5,2)	R_{52}	-4.799×10^{-2}	Path length from x' (dipoles)
$(l y)$	ME(5,3)	R_{53}	0	Path length from y
$(l b)$	ME(5,4)	R_{54}	0	Path length from y'
$(l l)$	ME(5,5)	R_{55}	1.000	Path length identity
$(l \delta)$	ME(5,6)	R_{56}	2.709×10^{-2}	Momentum compaction (27.09 mm)
(δx)	ME(6,1)	R_{61}	0	Energy from x
(δa)	ME(6,2)	R_{62}	0	Energy from x'
$(\delta \delta)$	ME(6,6)	R_{66}	1.000	Energy identity (implicit)
$(l \delta\delta)$	ME(5,66)	T_{566}	0	2nd-order momentum compaction

Table 2 summarises the beam parameters before and after transport. Table 3 shows the transverse emittance evolution.

Table 2: 6D bunch propagation results ($N = 10,000$). σ_δ is preserved exactly ($(\delta|x_j) = 0$). Bunch length grows due to $(l|\delta) \times \sigma_\delta$ chromatic path-length spread.

Scenario	σ_z^{in} (μm)	σ_z^{out} (μm)	$\sigma_\delta^{\text{in}}$ (%)	$\sigma_\delta^{\text{out}}$ (%)	σ_z ratio
0.5 ps, $h=0$	149.1	201.2	0.500	0.500	1.350
0.5 ps, $h=5\text{e}9$	149.1	255.0	0.557	0.557	1.710
2 ps, $h=0$	596.4	611.1	0.500	0.500	1.025
2 ps, $h=5\text{e}9$	596.4	875.8	1.111	1.111	1.469

Table 3: Transverse emittance through the COSY map. Horizontal projected emittance grows due to $(x|\delta) \times \sigma_\delta$ chromatic dispersion; vertical emittance is preserved exactly.

Scenario	$\varepsilon_{x,n}^{\text{in}}$ (μm)	$\varepsilon_{x,n}^{\text{out}}$ (μm)	$\varepsilon_{y,n}^{\text{in}}$ (μm)	$\varepsilon_{y,n}^{\text{out}}$ (μm)
0.5 ps, $h=0$	8.07	9.04	8.07	8.07
0.5 ps, $h=5\text{e}9$	8.07	9.26	8.07	8.07
2 ps, $h=0$	8.07	9.04	8.07	8.07
2 ps, $h=5\text{e}9$	8.07	12.05	8.07	8.07

The 0.5 ps beam experiences 35% bunch lengthening ($\rightarrow 0.67$ ps); the 2 ps beam is relatively unaffected (2.5% growth). The absolute chromatic contribution ($R_{56}\sigma_\delta = 27.09 \text{ mm} \times 0.005 = 135 \mu\text{m}$) is the same for both, but it represents a larger fraction of the shorter initial σ_z . Energy spread is preserved exactly in all cases.

Figures 1 and 2 show the longitudinal phase space and bunch profiles.

4 $R_{56} = 0$ Objective Test

Table 4 compares the baseline (transverse-only) and augmented ($+(l|\delta) = 0$) COSY FIT results.

Table 4: Effect of adding $(l|\delta) = 0$ as a COSY FIT objective. The optimiser produces identical results.

Configuration	MSE	$(l \delta)$ (mm)	Max $ \Delta I $ (A)
Transverse only	2.27×10^{-7}	27.0890	—
Transverse + $(l \delta)=0$	2.27×10^{-7}	27.0890	0.000 000

Adding $(l|\delta) = 0$ has zero effect: identical solution, identical currents, identical R_{56} . Verification with extreme weights ($w = 100$ and $w = 10,000$) confirms that the Stage 11 quadrupoles have negligible leverage over R_{56} : at $w = 10,000$, the best reduction is only $9 \mu\text{m}$ (0.03%) while transverse MSE degrades by $2300\times$. The R_{56} is geometry-locked by the dipole angles and spacings.

5 Bunch Compression Feasibility

5.1 Analytical Framework

With linear chirp h (s^{-1}) and momentum compaction R_{56} :

$$C(h) = \frac{1}{|1 + h R_{56}/(\beta c)|}, \quad h_{\text{opt}} = -\frac{\beta c}{R_{56}} \quad (1)$$

The output bunch length including uncorrelated energy spread is:

$$\sigma_{z,\text{out}}^2 = (1 + h R_{56}/\beta c)^2 \sigma_{z,\text{in}}^2 + R_{56}^2 \sigma_{\delta,0}^2 + R_{51}^2 \sigma_x^2 + R_{52}^2 \sigma_{x'}^2 \quad (2)$$

The compression floor at $h = h_{\text{opt}}$ is:

$$\sigma_{z,\text{min}} \approx R_{56} \sigma_{\delta,0} = 27.09 \text{ mm} \times 0.005 = 135 \mu\text{m} \approx 0.45 \text{ ps} \quad (3)$$

5.2 Part A: Chirp Sweep

Using the W9 COSY transfer map:

- $h_{\text{opt}} = -1.107 \times 10^{10} \text{ s}^{-1}$ ($\varphi = 38.1^\circ$ off-crest)
- $h_{C=4} = -8.3 \times 10^9 \text{ s}^{-1}$ ($\varphi = 27.5^\circ$ off-crest)
- Compression floor: $\approx 0.45 \text{ ps}$

At $C = 4$ chirp, the output is $\sim 0.67 \text{ ps}$ — better than 2 ps but not reaching the 0.5 ps target. At optimal chirp, the minimum is $\sim 0.45 \text{ ps}$, formally below 0.5 ps but requiring 38° off-crest with substantial energy spread growth.

Figures 4 and 5 show the compression factor and output bunch length vs chirp rate.

5.3 Part B: RF-Track Validation (Post-C7 Fix)

Six compression scenarios were tracked through the full beamline using the C7-fixed RF-Track adapter. Table 5 compares RF-Track particle tracking against COSY map propagation.

RF-Track and the COSY map disagree significantly for the chirped scenarios. At $C = 4$ chirp (B3), RF-Track gives $\sigma_z = 1.94 \text{ ps}$ vs the COSY map's 0.67 ps — a $2.9\times$ discrepancy. Several factors contribute:

Table 5: RF-Track (post-C7) vs COSY map propagation for 6 compression scenarios. All beams start at 2 ps. RF-Track includes physical apertures (quad bore 27 mm, dipole gap 14.5 mm); COSY map is linear (no apertures).

Scenario	σ_E (%)	h (s ⁻¹)	σ_z RFT (ps)	σ_z MAP (ps)	C RFT	T (%)
B1: baseline	0.5	0	2.360	2.038	0.847	83.5
B2: $C=2$ chirp	0.5	-4.2×10^9	2.170	1.313	0.921	70.3
B3: $C=4$ chirp	0.5	-8.3×10^9	1.937	0.671	1.032	58.8
B4: $C=6$ chirp	0.5	-12.5×10^9	1.805	0.522	1.108	52.5
B5: $\sigma_E=2\%$	2.0	0	2.315	2.681	0.864	56.2
B6: $\sigma_E=3\%$	3.0	0	2.299	3.355	0.870	49.6

- **Aperture losses:** transmission drops from 83.5% (B1) to 52.5% (B4) as the chirp-induced energy spread increases the beam size at aperture-limiting elements. Lost particles (those with the largest δ) are preferentially the ones that would contribute most to compression.
- **Nonlinear effects:** RF-Track includes higher-order optics and chromatic aberrations absent from the linear COSY map. At large chirp ($\sigma_\delta > 1\%$), these become significant.
- **Model differences:** the analytical sector-bend correction in RF-Track omits the triangle-rule fringe correction, producing slightly different path-length accumulation.

For the unchirped high- σ_E scenarios (B5, B6), RF-Track shows *less* elongation than the COSY map, because aperture losses remove the extreme- δ tails that drive the $R_{56}\sigma_\delta$ term.

Figure 8 shows the comparison graphically.

5.4 Part C: Extended Quadrupole Bounds

Re-running the 11-stage optimisation with chromaticity quad bounds raised from 10 A to 15 A (5 restarts) improves the transverse MSE but leaves the longitudinal dynamics unchanged. R_{56} depends only on dipole geometry, not quad currents.

5.5 T_{566} Assessment

$T_{566} = 0$ for the UH FEL transport line (Table 1). There is no nonlinear compression or decompression effect. A T_{566} FIT objective would be redundant.

6 Discussion

6.1 Bunch Lengthening Mechanism

The dominant source is $(l|\delta) \times \sigma_\delta$: chromatic path-length spread converts energy spread into longitudinal position spread. For the unchirped beam:

$$\sigma_{z,\text{out}}^2 = \sigma_{z,\text{in}}^2 + R_{56}^2 \sigma_\delta^2 + R_{51}^2 \sigma_x^2 + R_{52}^2 \sigma_{x'}^2 \quad (4)$$

Numerically: $R_{56}\sigma_\delta = 135 \mu\text{m}$, compared with $R_{51}\sigma_x \approx 8 \mu\text{m}$ and $R_{52}\sigma_{x'} \approx 6 \mu\text{m}$. The chromatic term dominates.

For the chirped case, the additional stretching factor is $|1 + hR_{56}/\beta c|$.

6.2 Transverse Matching Independence

The $(\delta|x_j) = 0$ map elements confirm that no transverse coordinate affects the energy deviation. The transverse Twiss match — and all 23 quad currents — is mathematically independent of bunch length. This has been confirmed at four levels:

1. Analytic decoupling of the 6×6 transfer matrix (S9)
2. FELsim numerical verification at $\sigma_E = 0.5\%, 2\%, 3\%$ (S9)
3. COSY 6D DA map: $(\delta|x_j) = 0$ for all transverse x_j
4. 10^4 -particle propagation: energy spread preserved exactly (this report)

6.3 Compression Limits

The transport line has positive $R_{56} = 27.09$ mm, enabling compression via negative chirp. However:

- The compression floor (0.45 ps) is set by $R_{56} \times \sigma_{\delta,0}$ and cannot be reduced by quadrupole adjustments.
- At $C = 4$ chirp, the COSY map predicts 0.67 ps but RF-Track gives 1.94 ps with only 59% transmission. Aperture losses and nonlinear effects severely limit practical compression.
- Reducing R_{56} to zero (or negative) requires changing dipole angles or spacings — a hardware modification.

The transport line is a *beam transport system*, not a *bunch compressor*.

6.4 Comparison with Worldwide FEL Practice

The finding that transport line optics are independent of bunch length is consistent with standard practice:

- **LCLS:** Pulse length adjustments via linac RF phases, without transport re-optimisation [6].
- **SACLA:** Two beamlines simultaneously receive different bunch lengths via RF phase modulation [4].
- **FERMI:** Compression factor varied while keeping chicane geometry constant [3].
- **SPARC:** Velocity bunching demonstrated compression ratios up to $C = 14$ [2].
- **CLARA:** Designed for dual-mode operation using the same S-band linac [5].

6.5 Recommended Switching Procedure

The $2\text{ ps} \rightarrow 0.5\text{ ps}$ switch requires changing *only* injector/RF settings. The recommended approach is velocity bunching: shifting the first linac section phase to $\sim 30\text{--}60^\circ$ off-crest. All transport line settings (23 quad currents) remain identical.

7 Summary

1. **Transverse matching is independent of bunch length.** The $(\delta|x_j) = 0$ map structure confirms analytical decoupling. All 23 quad currents are identical for 0.5 ps and 2 ps modes.
2. $R_{56} = 27.09$ mm, $T_{566} = 0$. The non-zero R_{56} causes bunch lengthening (35% for 0.5 ps, 2.5% for 2 ps at $\sigma_\delta = 0.5\%$) but does not affect transverse matching.
3. R_{56} is **geometry-locked**. Adding $(l|\delta) = 0$ as a FIT objective has zero effect on the optimised solution.
4. **In-line compression is possible but limited.** The compression floor is ~ 0.45 ps. At $C = 4$ chirp: COSY map predicts 0.67 ps; RF-Track gives 1.94 ps with 59% transmission. Aperture losses and nonlinear effects degrade practical compression.
5. **RF-Track and COSY map disagree on compression.** The discrepancy grows with chirp, reaching $2.9\times$ at $C = 4$. Aperture losses, higher-order optics, and model differences all contribute (Table 5).
6. **Extended quad bounds** do not affect σ_z .
7. **Recommended path:** deliver pre-compressed beam via upstream velocity bunching. The transport line preserves whatever bunch length is delivered.

Figures

References

- [1] M. Weinberg, A. Fisher, and R. Li, “Design of the UH Mark V Free Electron Laser,” arXiv:2510.14061v1, Table I.
- [2] M. Ferrario *et al.*, “Experimental Demonstration of Emittance Compensation with Velocity Bunching,” Phys. Rev. Lett. **104**, 054801 (2010).
- [3] S. Di Mitri *et al.*, “Coherent synchrotron radiation and microbunching in bunch compressors and free-electron lasers,” Phys. Rev. ST Accel. Beams **15**, 020701 (2012).
- [4] K. Tono, T. Hara, M. Yabashi, and H. Tanaka, “Multiple-beamline operation of SACLA,” J. Synchrotron Rad. **26**, 595–602 (2019).
- [5] E. W. Snedden *et al.*, “Specification and design for full energy beam exploitation of the compact linear accelerator for research and applications,” Phys. Rev. Accel. Beams **27**, 041602 (2024).
- [6] SLAC National Accelerator Laboratory, “LCLS Machine FAQ,” <https://lcls.slac.stanford.edu/machine/faq>.

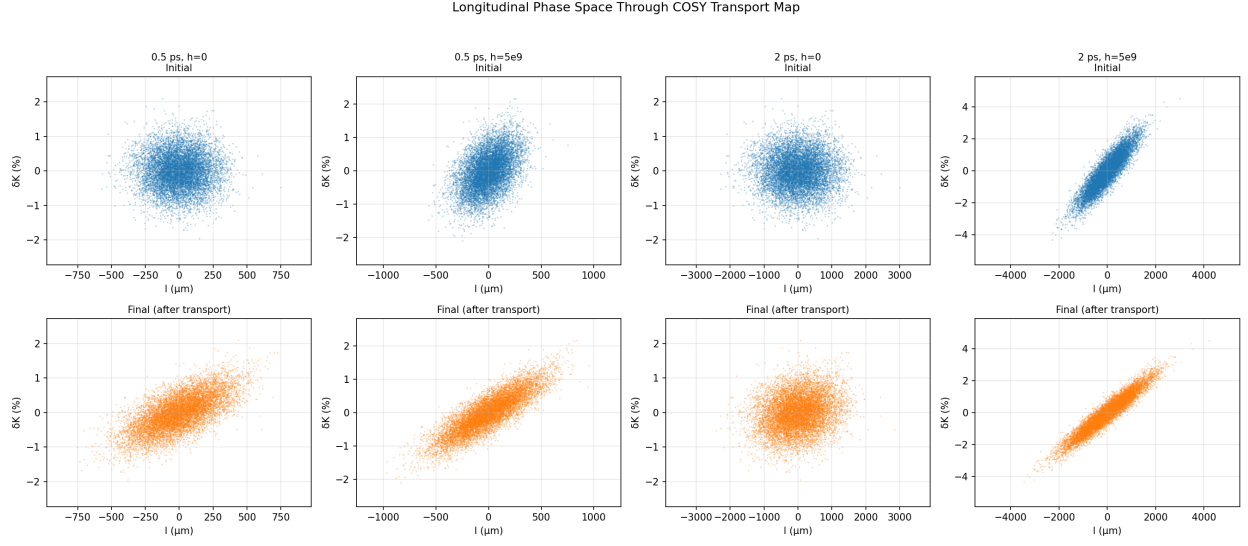


Figure 1: Longitudinal phase space (l vs δ) before (top) and after (bottom) transport through the COSY map. The 0.5ps distributions broaden visibly in l ; the 2ps distributions are largely unchanged. Energy spread is preserved in all cases.

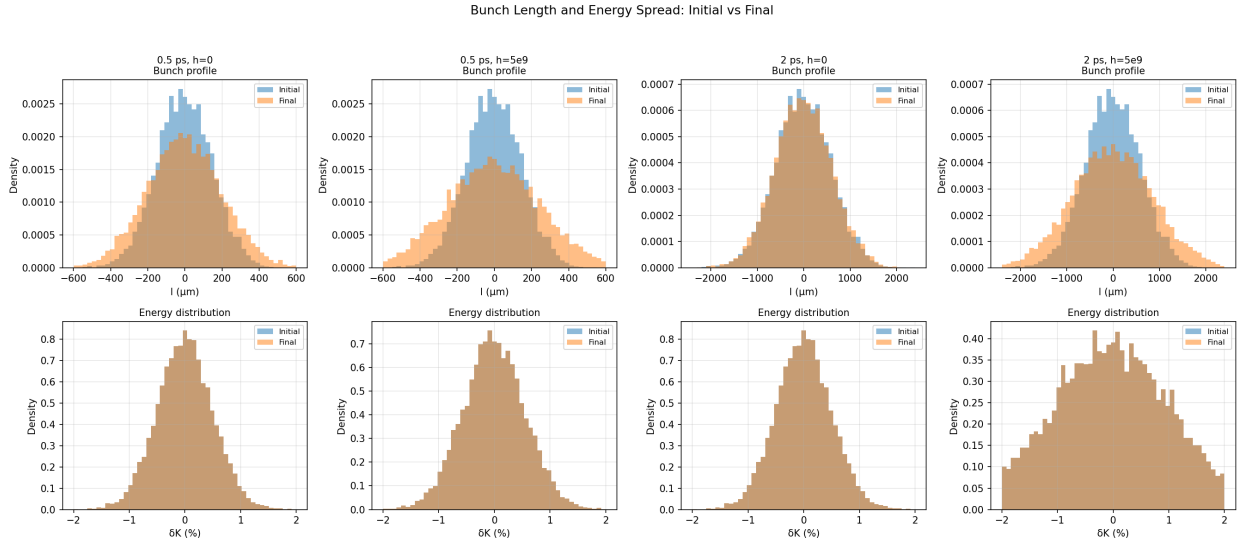


Figure 2: Bunch length (top) and energy distribution (bottom) histograms: initial (blue) vs final (orange). Energy distributions overlap exactly; bunch profiles broaden for 0.5ps due to chromatic path-length spread.

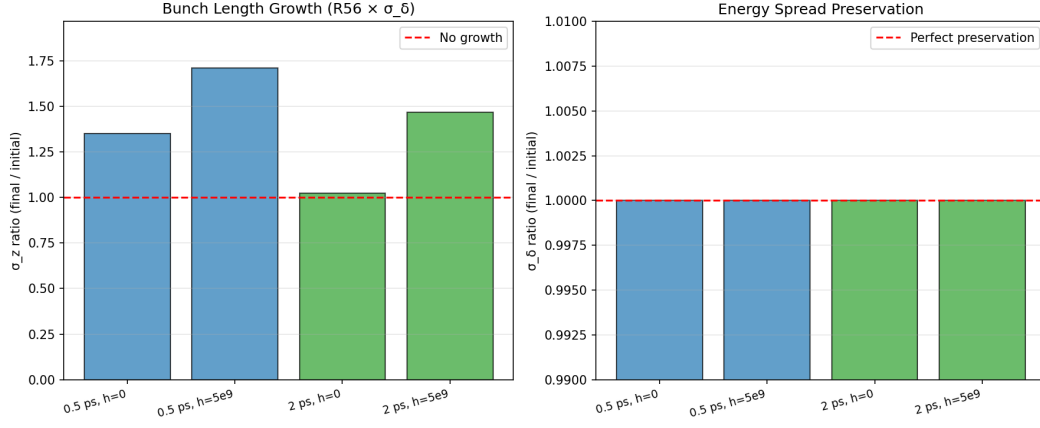


Figure 3: Left: bunch length growth ratio $\sigma_z^{\text{out}}/\sigma_z^{\text{in}}$. Right: energy spread preservation (ratio = 1.0000 for all scenarios).

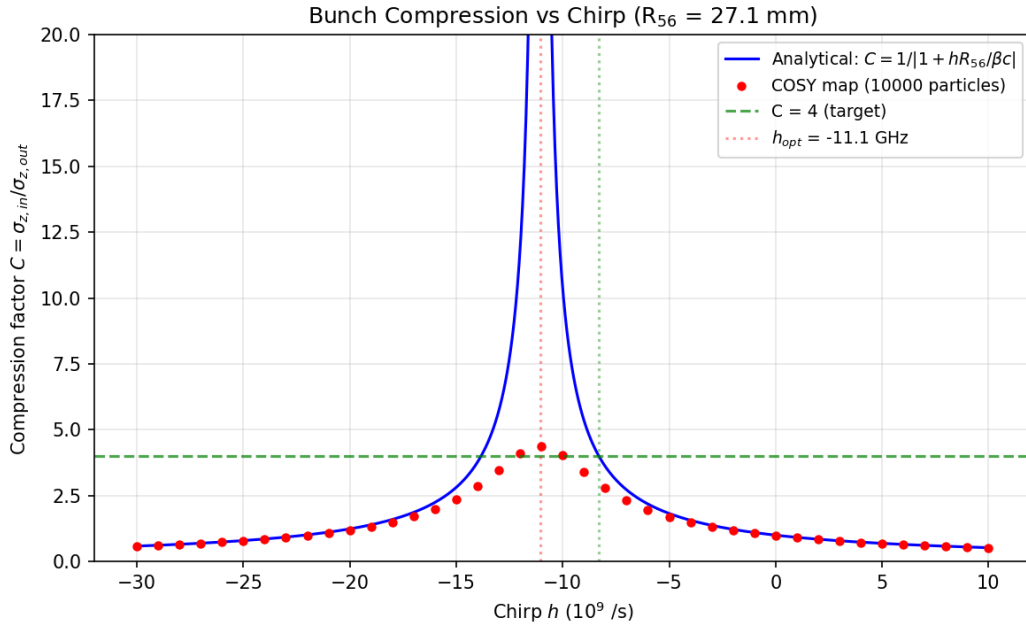


Figure 4: Compression factor vs chirp rate. Blue line: analytical $C = 1/|1 + hR_{56}/\beta c|$. Red circles: COSY map propagation. Green dashed: $C = 4$ target.

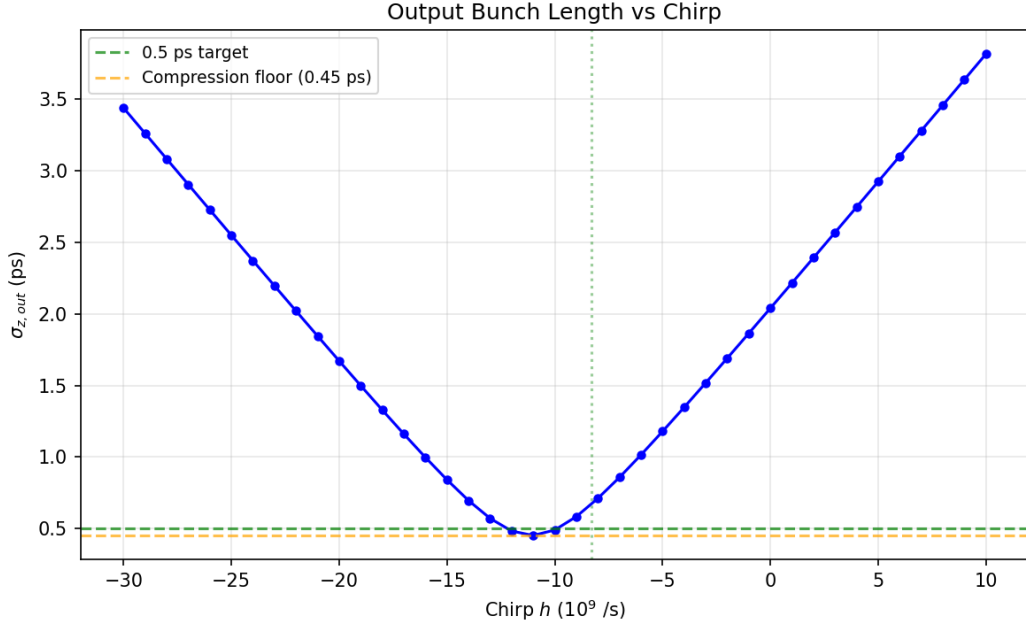


Figure 5: Output bunch length vs chirp rate. The compression floor (orange dashed, 0.45 ps) prevents reaching 0.5 ps (green dashed) at practical chirp values.

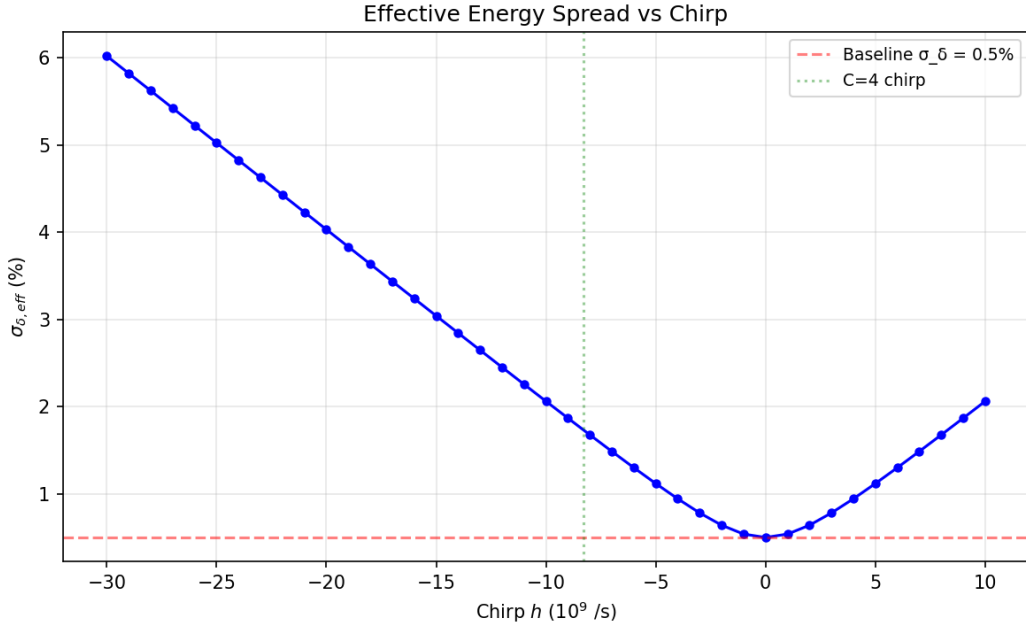


Figure 6: Effective energy spread vs chirp rate. At $C = 4$ chirp, $\sigma_{\delta,eff} \approx 1.7\%$.

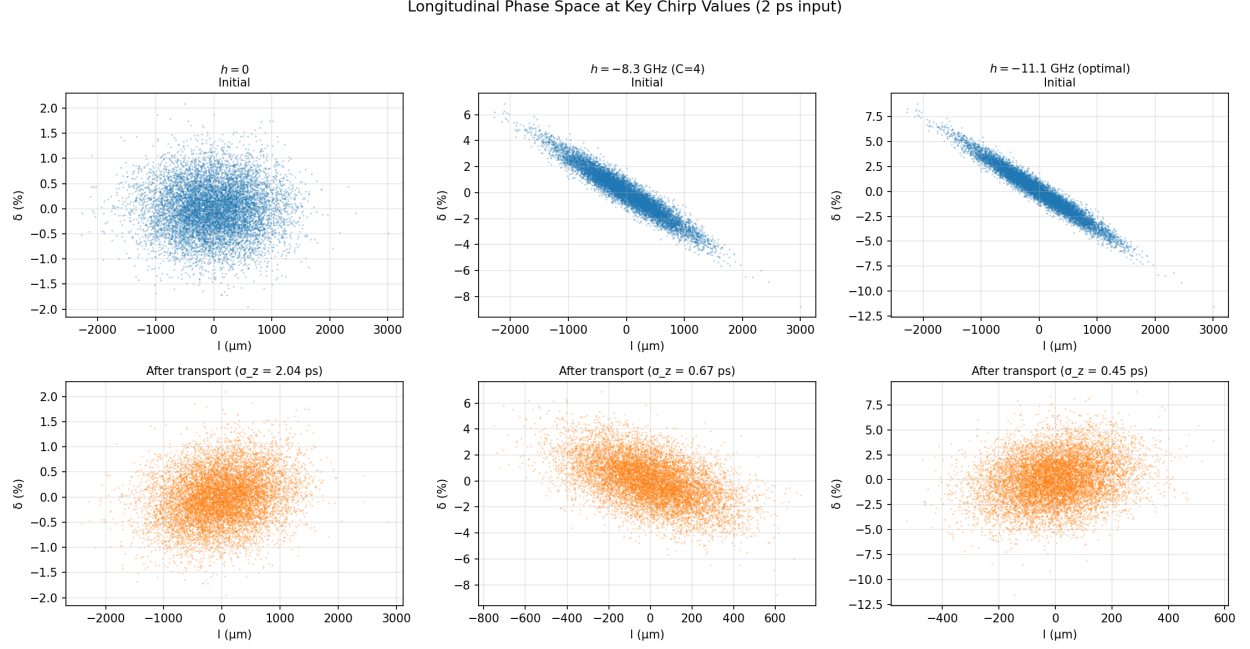


Figure 7: Longitudinal phase space before (top) and after (bottom) transport at three chirp values. At h_{opt} : maximum compression to the floor.

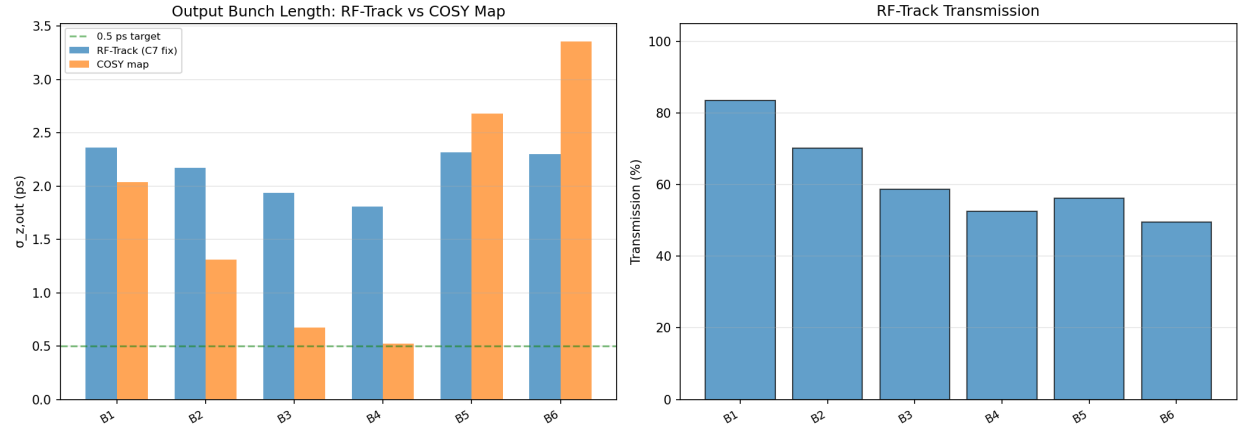


Figure 8: Left: output bunch length from RF-Track (blue) vs COSY map (orange) for 6 compression scenarios. Right: RF-Track transmission. The disagreement grows with chirp (B3–B4) due to aperture losses and nonlinear effects.